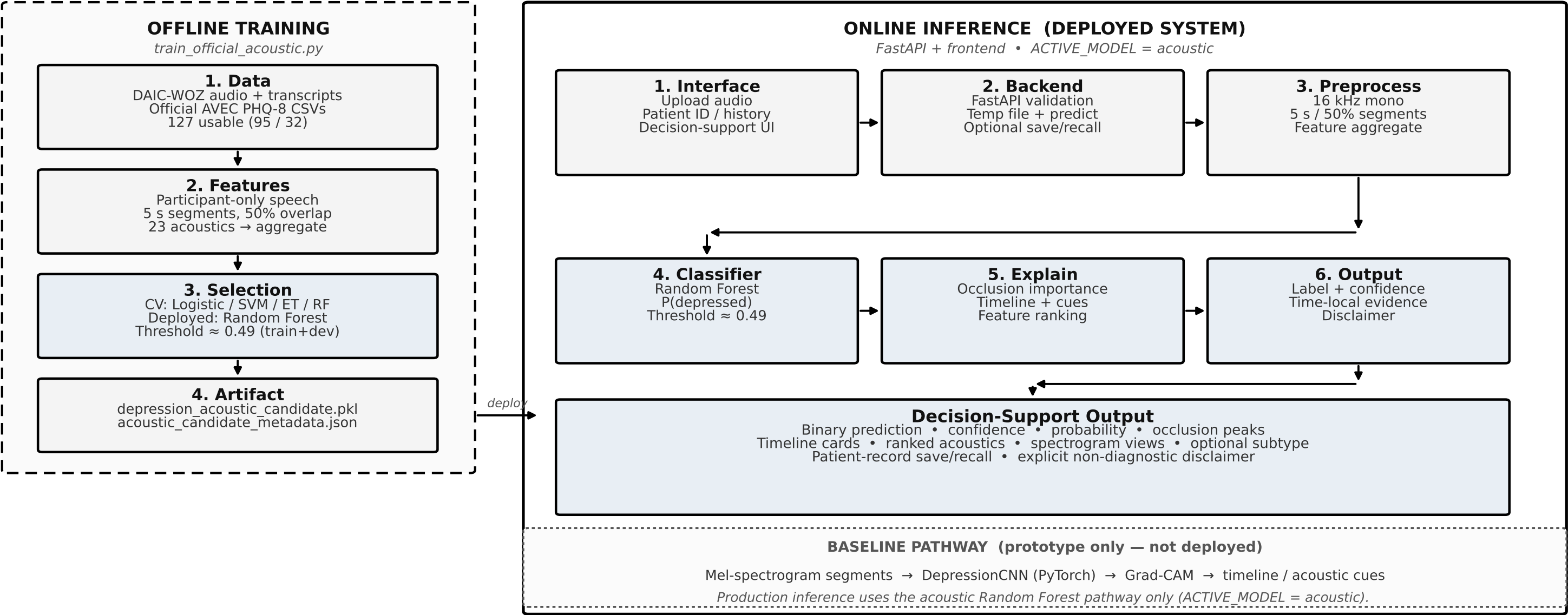


Figure 5.1 Top-Level System Architecture
Explainable Speech-Based Depression Detection Framework



OFFICIAL-LABEL EVALUATION (PRIMARY RESULT)
Labels: Official PHQ-8 binary • Split: AVEC train+dev (95) / held-out test (32) • Threshold ≈ 0.49
Deployed Random Forest held-out: Acc. 75.0% • Bal. Acc. 72.3% • Recall 63.6% • Spec. 81.0% • F1 63.6% • ROC-AUC 0.654
Confusion matrix: TN=17, FP=4, FN=4, TP=7 • Bootstrap Acc. 95% CI ≈ 59.4% – 87.5% • Decision-support prototype (not diagnosis)
Previous Extra Trees candidate (not deployed): Acc. 59.4%, recall 63.6%.

- Legend
- Deployed / key component

Supporting component

Offline training

Baseline only

Note: Deployed system = Random Forest acoustic pathway + occlusion explanations. CNN + Grad-CAM = baseline prototype only.